

Multivariate statistics

Lecture 11:

Classification with several populations

Literature and Problems: See course schedule

Agenda (preliminary):

Introduction

- **SLIDE: several-populations classification model and definitions**
- **DEMO: three-population classification, case I and II**

General classification criteria (distribution free)

- ECM, Expected Cost of Misclassification, definition
- minimizing ECM ---> splitting of sample space ---> decision rule
- special case: equal costs ~ minimum TPM, Total Prob. of misclassification
- special case: equal costs and priors ~ ML-rule
- MAP, Maximum Aposteriori rule: Bayes rule, posteriors ---> MAP-rule

Classification for several MVN populations, general case (QDA)

- model with equal costs but unequal population covariance matrices
- QDA, Quadratic Discriminant Analysis decision rule (population version)
- sample based QDA decision rule

Classification for several MVN populations, special case (LDA)

- model with equal costs and equal population covariance matrices
- LDA, Quadratic Discriminant Analysis decision rule (population version)
- sample based LDA decision rule

MVN model control

- test for MVN samples
- Bartlett test for equal population covariance matrices
- MANOVA test for significant separation of populations

DEMO: three-population classification (QDA and LDA case)

Performance of classification ~ probability of misclassification

- Empirical APER, Apparent Error Rate, using training data confusion matrix
- Estimated AER, Actual Error Rate, using training data with cross-validation

DEMO: three-population classification, performance of QDA and LDA cases

SLIDE: Summarizing several-population classification procedures